Cognitive Evolution Engine (CEE) A Middleware Framework for Self-Evolving Physical AI Robots Using Reflective Knowledge Intelligence

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***Abstract*— Autonomous robots improve their capabilities mostly via machine learning retraining and the collection of huge datasets. Still, they do not possess the ability to reason about failures, utilize experience, and adjust to changing conditions. The proposed Cognitive Evolution Engine (CEE) is a middleware designed to enhance the evolution of robot's knowledge through experience and reflective reasoning instead of continuous model retraining. This system operates at the junction of perception and decision-making, translating perceptual observations into operational knowledge with the help of seven components: Experience Interpreter, Reflective Reasoning Engine, Operational Knowledge Graph, Knowledge Evolution Engine, Intelligent Forgetting Module, Curiosity Engine, and Knowledge Transfer Engine. CEE is designed for mobile robots and works based on ROS 2. The ability to maintain constant knowledge acquisition, updating, and use of the knowledge allows CEE to avoid the need for continual retraining of its algorithms. Warehouse navigation demonstrates CEE's efficiency in learning from navigation failures, improving its knowledge base and applying it in future situations.**

Keywords— Autonomous Robots, Physical AI, Cognitive Middleware, Reflective Reasoning, knowledge Graph, Lifelong Learning, Explainable AI, ROS 2.

I. INTRODUCTION

The emergence of autonomously operating robots that are capable of functioning effectively under ever-changing and complex conditions has gained prominence in logistics, manufacturing, inspection, health care, and service areas. Autonomous operation implies that robots must adapt to new conditions as well as take advantage of the acquired information from previous interactions. As a result, continual and lifelong learning practices are recognized as effective approaches to improving robotic autonomy [1], [2]. Moreover, cognitive robotic schemes have placed more emphasis on memory, learning, reasoning, and knowledge retention in providing long-term autonomous operation capabilities [3].

Nevertheless, existing robotic learning systems are still mainly concerned with the performance of specific operations or the execution of a given model and methodology. In traditional robotic pipelines, the obtained measurement data is used for prediction or reaction purposes, while the gained experience remains unrecorded and is not preserved within the robotic system. This limitation is critical when considering prolonged use of robotic systems where robots operate in different environments and situations. Consequently, robots should be capable of interpreting past experience, keeping relevant operational information, and applying the provided knowledge during further work.

Recent research has been done on continual learning, lifetime robotics learning, robot memory, knowledge identification, reflective planning, experiments, and knowledge transfer [1]–[17]. These methods can provide the important capabilities for robotic intelligence with persistent functionality, but usually it is analyzed separately or in task-related systems. The integration of experience into more flexible operating in robots is still the theme under research.

The article introduces the concept of the Cognitive Evolution Engine (CEE), which is defined as a cognitive middleware system meant for experience-driven adaptation in physical AI robots. CEE is situated between perception and decision-making and is created for the analysis of sensory information and results of actions done by a robot, identification of important events, generation of operational knowledge, and preservation of this knowledge for future operations. The suggested architecture combines the elements of experience generation, reflective reasoning, operational knowledge identification, evolution, and reuse in one cognitive layer.

The functional CEE prototype has been constructed by utilizing an interactive mobile-robot simulation environment, a telemetry communication system based on MQTT, a rulebased cognitive reasoning module, a knowledge base, and a real-time visualization interface. The multi-variable

telemetry is utilized for modeling different states, such as wheel slip, obstacle proximity, collision, traction loss, and steep incline, and to formulate appropriate operational rules for solving the problems.

The main contributions of the work include:

Cognitive middleware architecture: The cognitive architecture is introduced in-between the robotic perception layer and the layer of decision-making.

Knowledge-rights based experience: The developed rulebased cognitive system transfers interpreted robotic state into a corresponding operational rule system

Memory: The generated operational rules are saved for future robotic use.

Implementation on a mobile robot prototype: The proposed strategy was tested within the framework of a mobile robot simulation and data transmission system based on MQTT.

II. Related Works

Long-term and continuous education are now considered to be very useful techniques that allow robots to gain new knowledge while keeping all the previously acquired skills through sequential tasks. Lesort et al. proposed a framework to achieve continuous learning in robotics, focusing on changing data distributions, catastrophic amnesia, storage and computing limits as well as real-world robotic needs [1]. Romero et al. researched long-term open-ended education from the point of view of cognitive structures that allowed them to determine contextual long-term memory, methods of learning, motivation, attention, and knowledge retrieval as the crucial elements of long-term autonomous existence of robots [2]. Subsequent studies considered the issue of continuous learning in physical robotics. Ayub et al. combined such methods as continuous learning, interactive machine learning, semantic reasoning, object recognition, and short- and long-term memory into one system for home-service robots [3]. Long et al. suggested DRAE, which includes dynamic specialist selection, retriever-augmented generation, hierarchical reinforcement learning, and long-term knowledge accumulation methods [4]. Meng et al. studied long-term reinforcement learning focused on using and reusing the information obtained in the process of completing various types of manipulative tasks [5]. Most recently, the success of CLARE has been notable in lightweight continual adaptation of vision-languageaction model under perspective of catastrophic forgetting and cost of the model updating [6]. These approaches resulted in efficient adaptation and reuse of knowledge; however, the approaches were mainly concerned with the achievement at the level of models, policies, or skills.

Persistent memory is considered another area of research concerning long-lasting robotic autonomy. Lei et al. suggested RoboMemory, which, in a single system, brings together spatial, temporal, episodic, and semantic memory, closedloop planning techniques, and dynamic knowledge graph techniques with respect to lifelong learning [7]. Beetz et al. introduced CRAM 2.0, designed in such a way that it aggregates semantic and episodic memory, generalized action plans, perception, constraint-based motion planning, and metacognitive reasoning [8]. The systems in question demonstrate the importance of memory keeping in the context of our interactions; however, mining memory does not automatically mean that physical phenomena observed earlier can be transformed into operational knowledge.

Knowledge representation and retrieval have been researched to assist robotic planning and execution. Vorobiov et al. created a knowledge-based robotic system that acquires motion primitives, action templates, safety rules, and information to facilitate an organized action planning process [9]. Shek et al. launched KGLAMP upon the merging of relationship, property, and reachability graphs with language-model-based planning for adaptive multi-robot planning and re-planning [10]. Kumpel et al. studied actionable knowledge graphs that¨ can relate instructions in natural language with robotic actions that can be repeatedly used [11]. The findings of these studies lead to the conclusion of the importance of having structured knowledge necessary for robotic reasoning and action selection. However, most of the knowledge obtained comes from the materials that are already available, demonstrations, or the information that has to do with the specific area of the query rather than the one that evolves from the individual experience of the robot.

Reflection and closed-loop reasoning have become widely discussed in recent years as a means of improving robotic adaptation. Lou et al. created ExploreVLM that combines exploration, object-based spatial representation, self-reflection, and execution verification [12]. Feng et al. created Reflective Planning that applies predicted future states. In their work, Feng et al. proposed Reflective Planning, a novel technique that assesses different future states and revises actions in longterm manipulation tasks [13]. In addition to Reflective Planning, Sun et al. presented a new method that effectively combines retrieval-augmented hierarchical reinforcement learning and hindsight modular reflection, which allows for the reuse of prior experiences in future task undertaking [14]. These proposed techniques exemplify that it is possible to improve planning and adjustment through reflection and experiencebased evaluation.

Autonomous exploration and knowledge transfer act synergistically with each other when it comes to lifelong learning. Fallas Hernandez et al. were the first to introduce a method´ for automatic sub-goal generation where they used intrinsic motivation, bottom-up learning, and linking of targets to set effective learning goals [15]. Gao et al. presented Arcadia, which merges exploration, grounding, scene reconstruction, shared embodied representations, and simulation-from-real into one system for lifelong learning tasks [16]. Overall, Yao et al. were the first to propose Primitive Prompt Learning, which provides a very efficient way of knowledge transfer through motion-aware prompts and text [17]. All these works indicate the importance of both obtaining new experience and reusing what is already known.

In general, previous studies have demonstrated several important features of robotic autonomy, including ongoing adaptation, consistent memory, organized knowledge, introspection, investigating, and transferring knowledge. However, these characteristics are usually studied separately or within systems designed for certain tasks. Hence the need for a comprehensive cognitive middleware framework which will help organize the way experiences are understood and used in the development of robotic systems.

III. Proposed Methodology

The Cognitive Evolution Engine (CEE) presented here acts as a cognitive middleware layer that lies between the acquisition of robotic experiences and the processing of operational decisions. The prototype has been designed in a way that enables it to collect the robot's telemetry data, interpret its current condition, detect pre-defined operational events, create necessary operational rules accordingly, and then store the operational rules in the historical knowledge. The architecture of the system in its entirety is shown in Fig. 1.

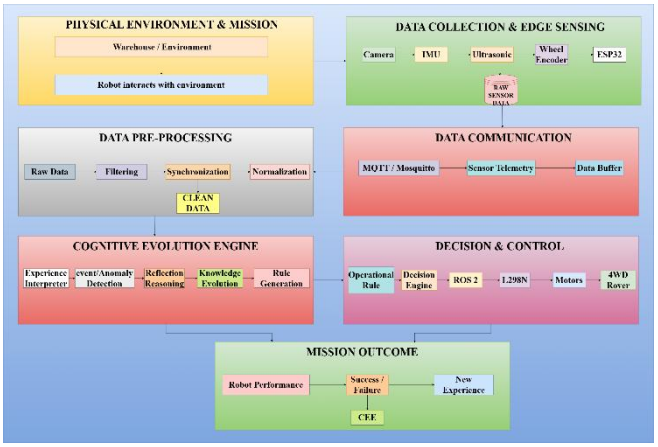


Fig. 1. Architecture of the Proposed Cognitive Evolution Engine (CEE) Framework

A. CEE Architecture

The CEE architecture has six functional layers in total: (1) physical setting and purpose, (2) data collection and edge monitoring, (3) data transmission, (4) cognitive evolution mechanism, (5) knowledge base, and (6) decision and control interface.

The prototype allows testing of experience generation, communication, cognition, storing knowledge, and visualizing processes within an interactive mobile robot simulation system. The simulator is able to create various robotic experiences, combining levels of obstacles approach, collision, wheel slip, traction loss, tilt, etc. The telemetry information is forwarded to CEE with the help of MQTT protocol that connects all elements of CEE.

Robot condition is shown as follows:

$$x_t = [R_t, A_t, P_t, U_t, T_t]^T$$

In that equation, R_t means average wheel speed and A_t means IMU acceleration, while P_t is pitch angle, U_t is ultrasonic distance, and T_t is the temperature.

The telemetry is moving through the following path:

Robot Simulator→MQTT Publisher→MQTT Broker→CEE.

The telemetry message received is represented as :

$$M_t = \{R_t, A_t, P_t, U_t, T_t\}.$$

As a result, using MQTT allows separating telemetry generation from the CEE process.

B. Cognitive Rule Generation And Event Interpretation

Processing of CEE as a whole can be defined as Telemetry → State Interpretation → Event Detection → Rule Generation → Knowledge Retention

The prototype implements the use of deterministic, condition-based cognitive reasoning rather than neural networks or reinforcement learning, enabling it to produce clear links between the telemetric data and the created operational rules.

Interpreting states and detecting is presented as

$$I_t = f_I(\mathbf{x}_t),$$

$$E_t = f_A(\mathbf{x}_t)$$

I_t – interpreted operational state and E_t – detected event. Multiple telemetry variables are evaluated in parallel if needed.

For example, wheel slip can be detected by applying the rule,

$$R_t > 120 \wedge A_t < 0.8$$

It means that IF wheel slip is detected, reduce motor RPM by 50%. Obstacle proximity detection is performed using:

$$U_t < 25\text{cm} \wedge R_t > 50$$

C. The Method of Keeping Operational Knowledge and Latching Events

If an event has been detected, a corresponding operational rule is stored in CEE operational knowledge archives. The applied method is historical operational rule storage rather than semantic knowledge graph usage. As for a specific knowledge item, it is expressed as

$$K_i = (ID_i, E_i, R_i) \quad (2)$$

where ID_i is the identifier of the rule, E_i is the event being detected, and R_i is the operational rule. The knowledge state is refreshed according to the following process:

$$K_{t+1} = K_t \cup \{K_i\} \quad (3)$$

The process of continuous telemetry means that an occurrence can remain active for several cycles. To avoid a repeated insertion of the same operational rule, the prototype operates using observance of the anomaly-latching method.

$$L_t = \begin{cases} 1, & \text{if an anomaly is active} \\ 0, & \text{otherwise} \end{cases}$$

The generation of a new operational rule takes place only if

$$L_{t-1} = 0 \wedge L_t = 1 \quad (5)$$

when,

$$L_{t-1} = 1 \wedge L_t = 1 \quad (6)$$

Once active anomaly remains stable and does not create any knowledge entry duplicates.

D. Processing Flow and Visualization of Cognitive Events

The entire process for processing is shown in Fig. 2. As for the robot operation, telemetry is transmitted via MQTT and parsed by the CEE into its position in state. These conditions are to be evaluated. In case an event is detected, the so newly discovered state is checked.A newly linked event establishes an operational rule and adds it to the knowledge history. The transition thus made is summarized as

$$\mathbf{x}_t \rightarrow E_t \rightarrow R_t \rightarrow K_{t+1} \quad (7)$$

The CEE dashboard allows the monitoring of telemetry, system status, cognitive mapping, and operational knowledge history, permitting aspects such as the relationship between telemetry, detected event, and operational rules during operating time to be monitored.

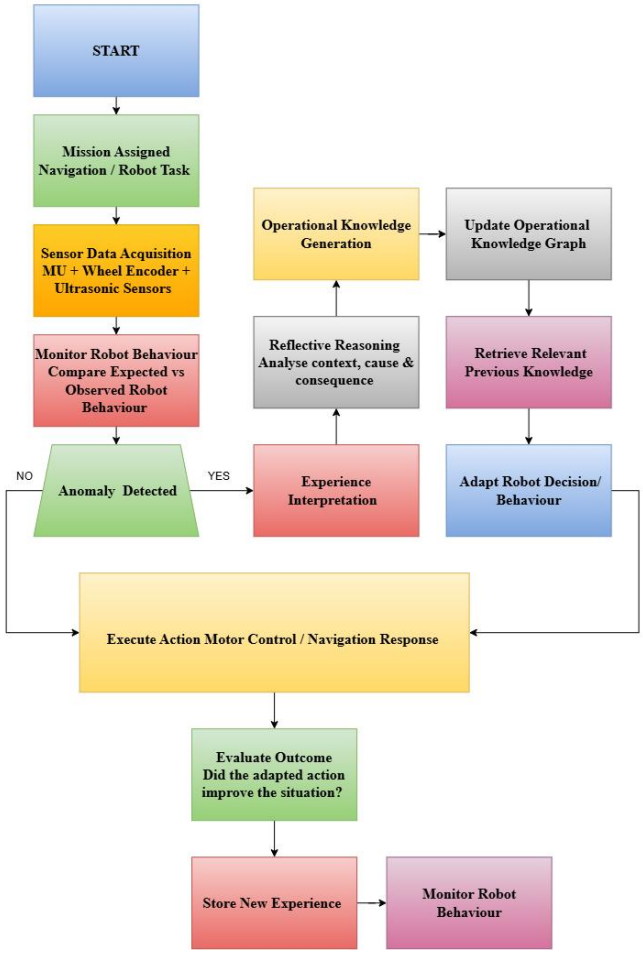


Fig 2. Processing flow of the implemented CEE

E. Experimental Procedure and Evaluation

The experimental procedure evaluates the ability of completely simulated robot operating conditions to be correctly comprehended and converted into the expected operational rules. The simulator and CEE interface are initialized with the establishment of the MQTT telemetry channel beforehand. The active scenarios are then run to consider their conditions of operations down to the last controllable scenarios. The accuracy of event detection is defined as

$$EDA = \frac{N_c}{N} \times 100$$

where N is the number of test events and N_c is the number of events that are correct. The generation rate of rules is defined as

$$RGR = \frac{N_r}{N_c} \times 100$$

where N_r is the number of correctly detected events for which the ideal rule has been generated. Knowledge retention rates are defined as:

$$KRR = \frac{N_k}{N_r} \times 100$$

where N_k is the number of rules that are successfully retained. Duplicate suppression rates are described as:

$$DSR = \left(1 - \frac{N_d}{N_r}\right) \times 100$$

where N_d is the number of events for which the rule has been generated many times. Cognitive response latency is established as:

$$T_c = T_r - T_e$$

where T_e is the time when the event has been received and T_r is the time at which the rule was generated. Telemetry processing rate is defined as:

$$TPS = \frac{N_m}{T}$$

where N_m is the number of telemetric messages that are processed in the given time interval T.

These indicators analyze the operation of the deterministic CEE pipeline without using the traditional machine-learning indicators that cannot be used in the current context of reasoning.

F. Configuration

The prototype comprises of an interactive mobile robotic environment simulator, the MQTT interface, the deterministic cognitive reasoning engine, knowledge log, anomaly-latching, and the real-time visualization dashboard.

1) TABLE I

CONFIGURATION OF THE PROTOTYPE

Component	Implementation
Robotic environment	Interactive mobile-robot simulator
Interaction	Manual keyboard control
Communication	Interface of MQTT
Cognitive process	Reasoning based on conditions
Event detection	Multi-variable thresholds
Rule construction	Relation between events and rules
Knowledge storage	Archive of the rules
Event control	Anomaly latching
Visualization	Real-time CEE dashboard

IV. RESULT AND DISCUSSION

The CEE prototype under discussion underwent tests in a mobile-robot simulation setting (a simulation environment). Systems were checked against requirements for telemetry use and decision-making. CEE prototype has identified

different processes of operation, in particular, such important conditions as telemetry failure, thermal overload, sensor blindness, and collision of objects. Every condition has been linked with an operational rule.

A. Development of Rules from Experience

Identification of the representative wheel-slip condition means

$$R_t > 120 \wedge A_t < 0.8 \tag{26}$$

Once the condition is satisfied, the CEE creates the following rule: Reduction of Motor RPM in case of Wheel Slip This is a sample indicating how the mapping between robotic observations and operational rules was performed.

B. Keeping Operational Knowledge

The generated rules are provided with a unique ID and stored in the operational repository.

$$K_{t+1} = K_t \cup \{K_i\} \tag{28}$$

In contrast to transient event display methods, the repository of rules is not emptied when the corresponding anomaly ceases because such methods perform the operation with transient events, while in this case, operational knowledge is extracted.

C. Visualization and Latching of Events

In order to avoid duplication of events during the ongoing anomalous situation, new rules are formed when the anomaly changes from inactive to active.

$$L_{t-1} = 0 \wedge L_t = 1 \tag{29}$$

The CEE is provided with a dashboard that offers visualization of telemetry information, detected events, created rules, and stored knowledge.

D. Discussion

The findings indicate the practicality of the Cognitive Evolution Engine as a mediating cognitive layer bridging robot telemetry and operational knowledge. The functioning device encapsulates in its processing logic the capability to receive MQTT) signals, interpret various events, formulate rules, retrieve knowledge, and visualize the result. Although the deterministic reasoning method allows establishing links between the conditions registered and rules generated, the results indicate the practical success of the prototype realized rather than general learning or autonomous knowledge growth.

V. CONCLUSION AND FUTURE WORK

The Cognitive Evolution Engine (CEE) is a cognitive middleware system for converting robot experiences into operational knowledge. It is executed in a functional prototype that includes a mobile robot simulation, MQTT telemetry transmission, cognitive processing, history of operational knowledge, event recording, and real-time visualization. This system processes wheel velocity, inertial data, pitch, distance measured by an ultrasonic device, temperature, etc., to determine operational conditions and create respective operation rules. The prototype displays the entire process from telemetry transmission to interpretation of events to rule creation and retention of knowledge. Evaluation metrics proposed here establish quantitative evaluation of accuracy of detection, rule creation, retention of knowledge, suppression of duplicates, speed of responses, and speed of telemetry processing. Future developments will include the generation of adaptive rules for cases that have not been seen before, the construction of semantic networks, intelligent forgetting, exploration prompted by curiosity, sharing of knowledge between robots, and testing of this system on a real robot with sensors and hardware.

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